STATISTICS

Statistics Cheat Sheet

The concepts interviewers ask, each explained the way you would say it to a manager.

- **Core concepts in plain words**
- **Which test to use, when**
- **A/B testing step by step**

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The basics

You will use these every week, not just in interviews.

- **Mean vs median**

  Use median when data is skewed. Average salary in a team with one CEO is misleading. Median is not.

- **Standard deviation**

  How spread out values are. Small SD means values sit close to the average.

- **68 · 95 · 99.7 rule**

  In a normal distribution, that share of values falls within 1, 2 and 3 standard deviations.

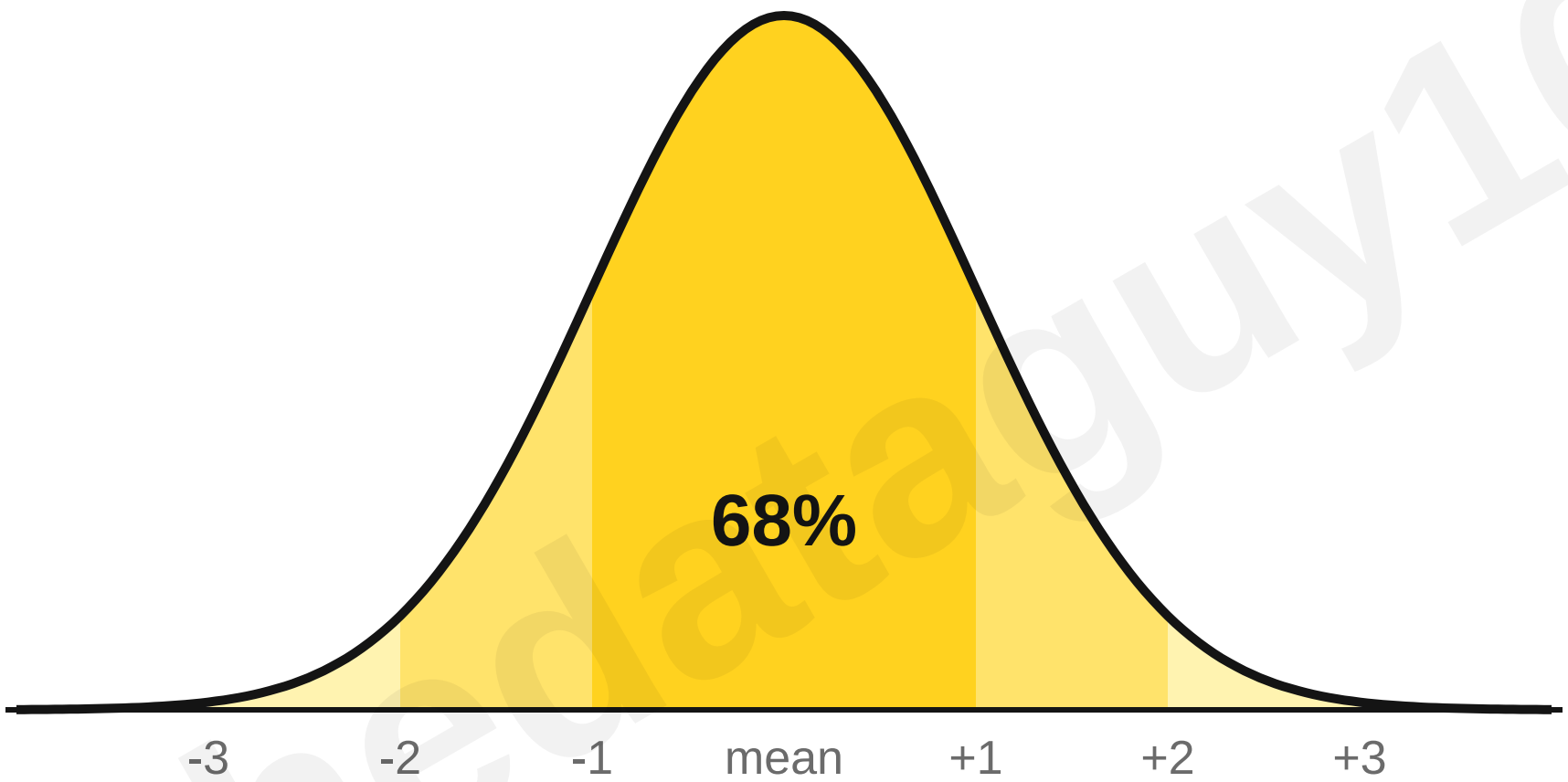
- **Outliers (IQR rule)**

  Below Q1 − 1.5×IQR or above Q3 + 1.5×IQR.

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The normal curve

Most measurements cluster around the average. This shape is the base of most tests.



95% within ±2 SD

99.7% within ±3 SD

Why it matters

A value beyond 2 SD happens about 5% of the time. That is the idea behind $p < 0.05$.

z-score

How many SDs a value is from the mean. $z = (x - \text{mean}) / SD$.

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Hypothesis testing

The most asked stats topic. Learn to explain p-value without jargon.

- **Null hypothesis**

  The default claim: nothing changed. The new button did not change conversion.

- **p-value**

  If nothing really changed, how likely is a result this big? Below 0.05 usually means significant.

- **Type I error**

  False alarm. You say the button worked when it did not.

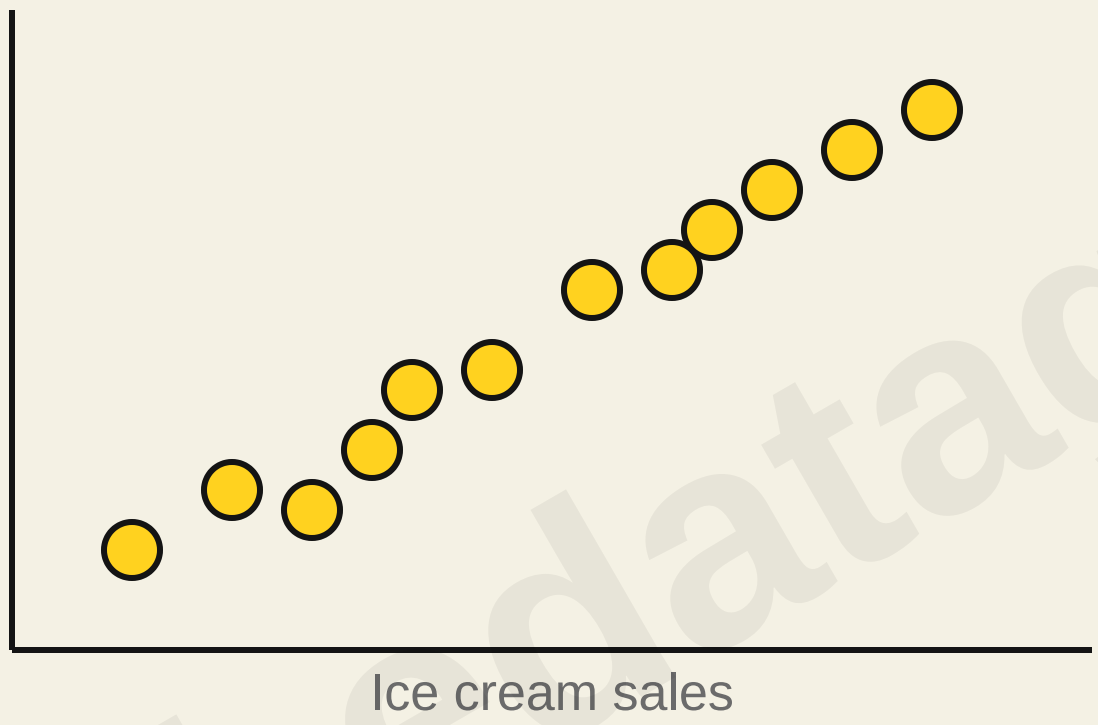
- **Type II error**

  Missed it. The button worked but your test did not detect it. Usually too small a sample.

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Correlation is not causation

The most common mistake in business analysis.



r = 0.92

with drownings.

Cause: summer hea

drives both.

Ask the third factor

What else could drive both numbers? Season, price, region?

Prove cause with a test

Only a controlled experiment (A/B test) shows cause.

Confidence interval

"Lift is 4% ± 1.5%" tells more than "lift is 4%".

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Which test to use

Pick by what you are comparing.

- **Two group averages**
  t-test. Average order value, app A vs app B.
- **Three or more averages**
  ANOVA. Average spend across 4 cities.
- **Two categories related?**
  Chi-square. Is gender related to product choice?
- **Two conversion rates**
  Two-proportion z-test. Old checkout 3.1% vs new 3.6%.
- **Two numbers related?**
  Correlation, then regression. Ad spend vs sales.

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A/B testing in 6 steps

Asked at every product company. Say the steps in this order.

1. One primary metric

For example checkout conversion rate.

2. Sample size first

Decide significance (0.05), power (80%) and the smallest lift you care about.

3. Random split

Users, not sessions, so one person sees one version.

4. Run full weeks

Weekday and weekend behaviour differ. Do not stop early when it looks good.

5-6. Test and check guardrails

Significant? Then make sure revenue or retention did not drop.

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10 questions with answers

Short, clear answers score better than textbook ones.

p-value to a manager

If nothing changed, a result this big would happen less than 5% of the time.

Median over mean?

When data is skewed or has outliers.

Significant but 0.1% lift?

Statistically real, but check if it is worth the cost.

Simpson's paradox

A trend in every group reverses when groups are combined.

Sample vs population

Sample is the part you measure. Population is everyone.

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